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Roll No

AU-701(A) (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Alternative Fuels and Pollution Control

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the renewable sources of energy for automobiles?
b) Discuss the factors affecting the selection of an alternative fuel.
2. a) Which is safer fuel between CNG and LPG?
b) Discuss the use of Ethanol for I.C. Engines.
3. Explain in brief the modification is to be required to use alternative fuels in SI and CI engines.
4. a) Explain the various types of solar-thermal collectors.
b) What is fuel cell? Explain its principle of working.
5. a) How CI engines can be used with vegetable oils as fuels?
b) How are can use hydrogen and biogas as fuel in IC engines?
6. State the classification of energy storage technologies used in automobiles. Discuss energy efficient batteries suitable for four-wheelers.

OR

- a) Describe the need of alternative fuels.
- b) Discuss the availability of biofuels in brief.
7. Explain the principle of conversion of solar energy into heat.

OR

What are the main applications of geothermal energy?

8. Write short notes on followings:(any two)
 - a) Smoke meters
 - b) European emission standards
 - c) Biodiesel as fuel
 - d) Catalytic converters
